



CONTACT



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EDUCATION

Bachelor of Technology (B.Tech)

Electrical Engineering
Bajaj Institute of Technology, Wardha
Expected Graduation: 2027
CGPA: 6.60

Higher Secondary Certificate (HSC)

Insight Junior College, Jamtha
Maharashtra State Board
Year: 2022
Percentage: 87.67%

Secondary School Certificate (SSC)

Kasturba Vidya Mandir, Sevagram
Maharashtra State Board
Year: 2020
Percentage: 85.20%

TECHNICAL SKILLS

Programming

- C
- C++
- Python
- Arduino IDE

Electrical Engineering

- Power Electronics
- Circuit Design
- Electrical Machines
- Basic Power Systems
- Industrial Automation

Embedded Systems

- ESP32
- Arduino
- Sensors & Actuators
- Relay & Motor Control

Design & Simulation

- AutoCAD Electrical
- MATLAB
- Simulink
- LTspice
- Tinkercad

Hardware

- Hardware Prototyping
- PCB Fundamentals
- Electronic Testing
- Soldering

KARAN KAILAS MENDHE

ELECTRICAL ENGINEERING UNDERGRADUATE | POWER ELECTRONICS | EMBEDDED SYSTEMS | ELECTRICAL DESIGN



ABOUT ME

Electrical Engineering undergraduate with a strong foundation in power electronics, embedded systems, industrial automation, and hardware prototyping. Experienced in developing real-world engineering projects involving ESP32, IoT, microcontrollers, and electronic systems. Passionate about designing practical engineering solutions through hands-on learning, innovation, and continuous skill development. A quick learner with strong problem-solving, teamwork, leadership, and communication skills, seeking opportunities to contribute and grow in the electrical engineering industry.



PROJECTS

1. Dextra – Smart Hand Orthosis (SIH 2025 Grand Finalist)

- Designed and developed a wearable smart hand orthosis using ESP32 and servo motors to assist wrist and finger rehabilitation.
- Implemented a tendon-driven mechanism for controlled finger actuation and precise hardware integration.
- Developed embedded control logic for smooth, reliable, and user-friendly operation.
- Contributed to hardware integration, prototype assembly, testing, and system validation.
- Optimized the prototype for lightweight construction, comfort, safety, and ease of use.
- Presented the project at the Smart India Hackathon 2025 Grand Finale as part of Team Dexteriest.

2. Smart Grid & Microgrid Monitoring System

- Developed an IoT-based monitoring system for real-time analysis of solar grid and microgrid parameters.
- Integrated voltage, current, power, temperature, and irradiance sensors with relay modules for data acquisition and automated control.
- Designed a real-time monitoring and data visualization system to analyze system performance and operational efficiency.
- Enhanced system reliability, safety, and energy management through intelligent monitoring and automation.
- Presented the project at Smartathon 2.0, a National Level Innovation Competition.

3. DC-DC High Boost Gain Converter

- Designed and fabricated a high-gain DC-DC boost converter for efficient voltage conversion.
- Performed hardware assembly, testing, and performance evaluation under different operating conditions.
- Improved converter efficiency and voltage gain through practical design optimization.

INTERNSHIP & TRAINING

Industrial Training – Electrical Loco Shed, Nagpur

Gained practical exposure to locomotive electrical systems, traction motors, battery banks, maintenance practices, and industrial safety procedures.

CERTIFICATIONS

MATLAB On ramp – MathWorks

Simulink On ramp – MathWorks

Introduction to Power Electronics – Coursera (University of Colorado Boulder)

Control Engineering – Coursera (University of Colorado Boulder)

ACHIEVEMENTS

- Smart India Hackathon 2025 – Grand Finalist (Team Dextra)
- Smartathon 2.0 – National Level Finalist
- Wardha Innovation Challenge – Runner-up
- DIPEX Innovation Competition – Regional Round Qualifier

PROFESSIONAL SKILLS

- Problem Solving
- Leadership
- Communication
- Teamwork
- Time Management
- Adaptability

AREAS OF INTEREST

- Power Electronics
- Embedded Systems
- Industrial Automation
- Electrical Design
- Renewable Energy
- Electric Vehicles

LANGUAGES

- English (Professional Working Proficiency)
- Hindi (Native/Bilingual Proficiency)
- Marathi (Native/Bilingual Proficiency)